Enrollment No.: _____

Darshan UNIVERSITY

Darshan Institute of Computer Application
BCA / B.Sc. (IT) | Sem-2 | Summer-2024

Course Code	:	2304CS204	**Date**	:	29-04-2024
Course Name	:	Database Management System - I	**Duration**	:	150 Minutes
			Total Marks	:	70

Instructions:
1. Attempt all the questions.
2. Figures to the right indicates maximum marks.
3. Make suitable assumptions wherever necessary.

Q.1 **(A)** Describe types of database users. **4**

(B) Define the following terms: **3**
(i) Data
(ii) Database
(iii) DBMS

OR

Define the following terms:
(i) Information
(ii) Metadata
(iii) Data Warehouse

(C) Write a short note on functions and responsibilities of DBA. **7**

OR

Write a short note on advantages of DBMS.

Q.2 **(A)** Compare Schema Vs Instance. **4**

(B) Discuss Hierarchical data model. **3**

OR

Discuss Object-Oriented data model.

(C) Describe mapping and its types with suitable diagram. **7**

OR

Draw and explain three level ANSI SPARC database architecture.

Q.3 **(A)** Explain UPDATE command with suitable example. **4**

(B) Explain RENAME command with suitable example. **3**

OR

Explain GROUP BY clause with suitable example.

(C) Prepare SQL queries for the below table: **7**

Movie (movie_id, title, release_year, director)

1. Display all the movie details of year 2021.
2. Display movie id and title of director 'Mahesh Bhatt'.
3. Delete all the movie where title is 'Hum Tum'.
4. Retrieve movie name starts with letter 'R' and having at least 6 characters.
5. Display all the movies release between year 2010 to 2018.
6. Retrieve all the movie whose id is 102, 105 and 111 (Use IN).
7. Add one more column Rating INT Null in movie table.

OR

Prepare SQL queries for the below tables:

City (<u>city_id</u>, city_name, pincode)

Village (village_id, village_name, city_id)

1. Retrieve all the city name of Baroda city.
2. Count total number of villages in each city.
3. Display unique city from city table.
4. Add one more column Remarks varchar(100) NULL to City table.
5. Display the city having no village.
6. Delete all the records of village table (Use Truncate).
7. Rename City table to City_Details.

Q.4 **(A)** Define the following terms: **4**

(i) Entity Set
(ii) Arity
(iii) Cardinality
(iv) Domain

(B) Explain UNION operation in relational algebra. **3**

OR

Explain INTERSECTION operation in relational algebra.

(C) Explain SELECTION and PROJECTION operation in relational algebra with suitable example. **7**

OR

Describe JOIN operation in relational algebra with suitable example.

Q.5 **(A)** Explain mapping cardinalities with suitable example. **4**

(B) Describe Simple attribute Vs Composite attribute with proper example. **3**

OR

Describe Single-valued Vs Multi-valued attribute with suitable example.

(C) Explain Specialization and Generalization with suitable diagram. **7**

OR

Draw and explain E-R diagram symbols.

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